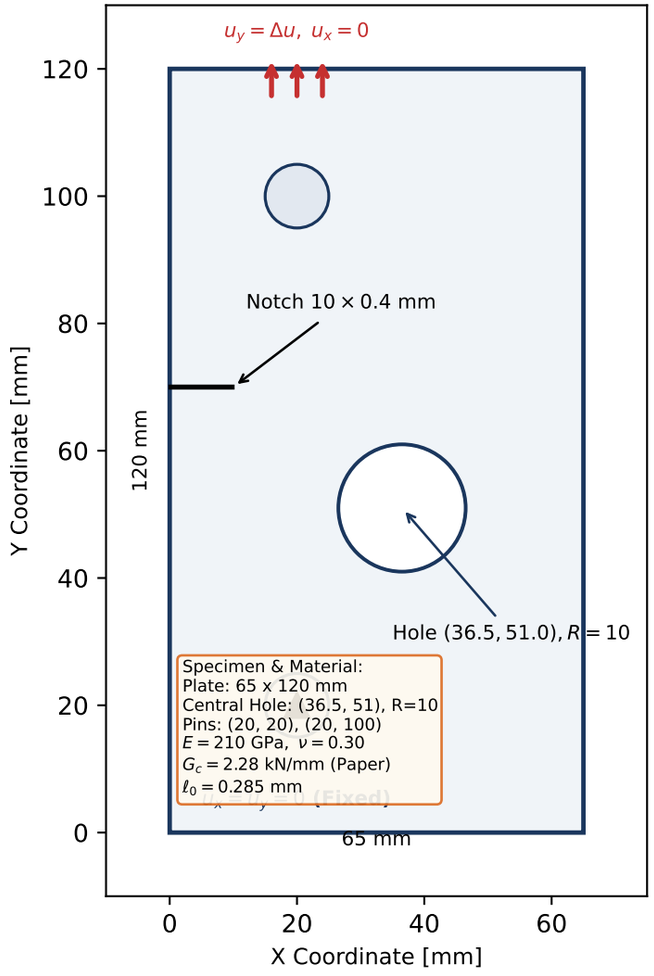
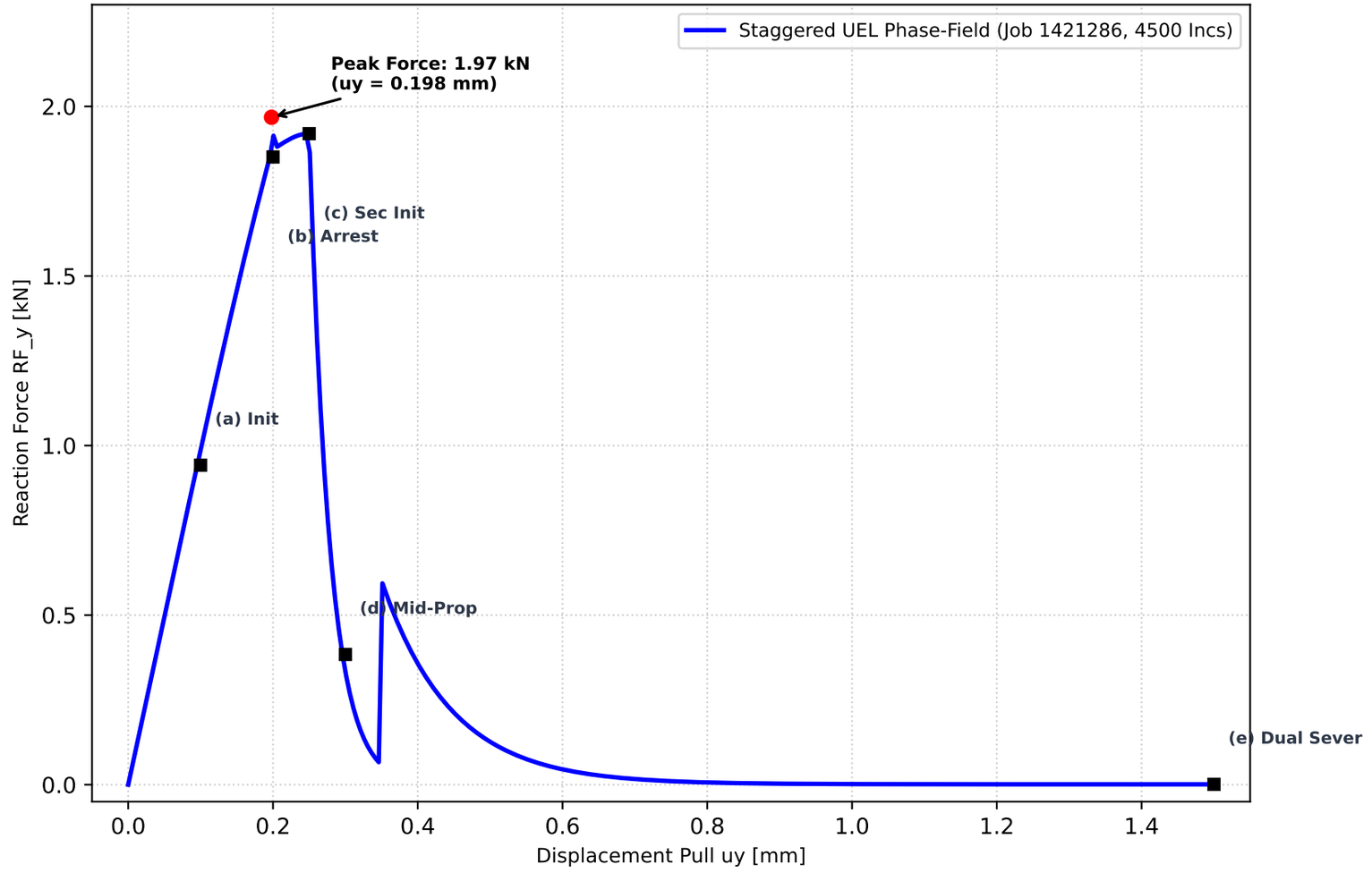
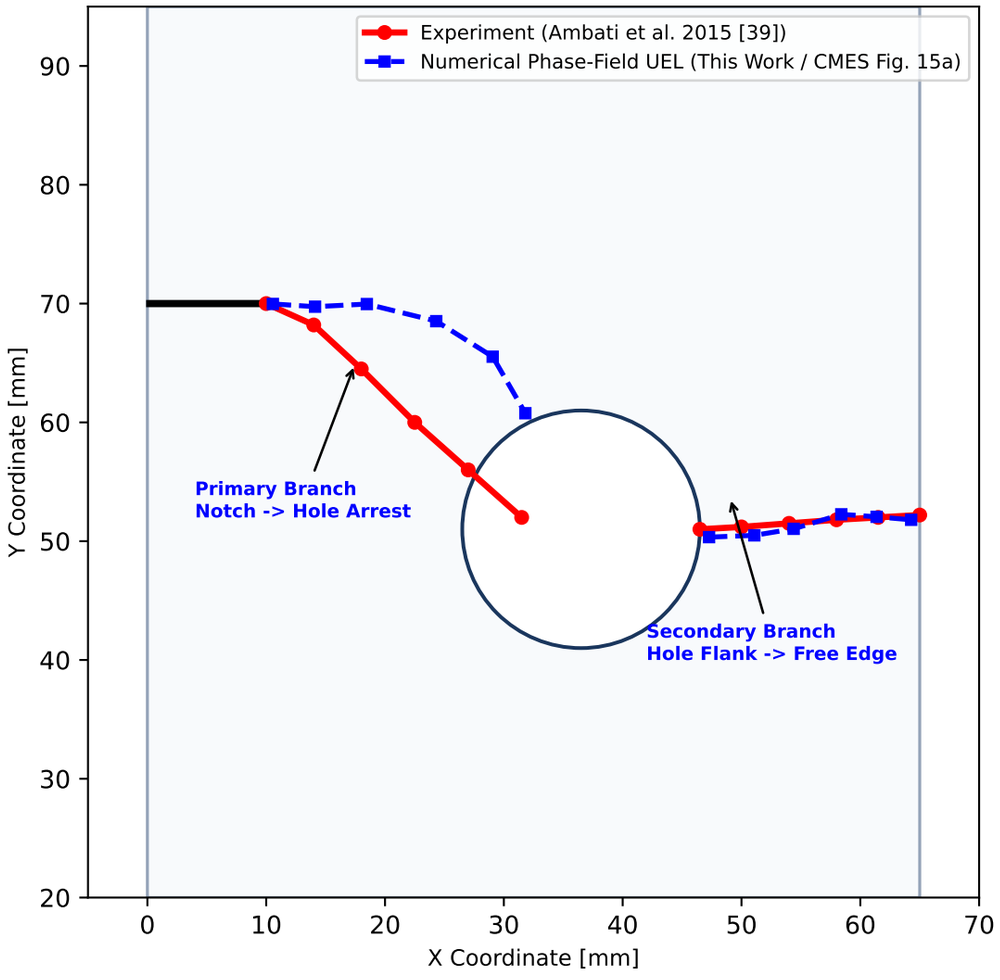
(a) Figure 14: Notched Holed Plate Geometry & BCs
Exact 65 x 120 mm Plate with Central Void and Initial Notch



(b) Figure 17(a): Structural Load-Displacement Response
Staggered UEL Phase-Field Simulation Capturing Dual-Crack Sequence



(c) Figure 15(a): Dual-Crack Trajectory Comparison
Excellent Agreement with Ambati et al. (2015) Experimental Paths



(d) Figure 16(b) Synthesis: Two-Branch Adapted Mesh
18,984 Elements (h_min <= l0/5) with Preserved Primary & Secondary Bands

